1.

Minimize the sum of squared residuals:

SSR = $\sum_{i=1}^{n}[y_i - (\alpha x_i + \beta)]^2$

Setting the partial derivatives with respect to α and β to zero gives:

$\hat{\alpha} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$

$\hat{\beta} = \bar{y} - \hat{\alpha}\bar{x}$

2.

From the given matrix: TP = 40, FP = 5, FN = 10, TN = 45

TPR = $\frac{TP}{TP+FN}$ = 0.8, TNR = $\frac{TN}{TN+FP}$ = 0.9

FPR = $\frac{FP}{FP+TN}$ = 0.1, FNR = $\frac{FN}{FN+TP}$ = 0.2

Recall = 0.8, Precision = $\frac{40}{40+5}$ = 0.8889

Accuracy = $\frac{40+45}{100}$ = 0.85, F1 = $\frac{2PR}{P+R}$ = 0.8421

3.

For query q = (3.5, 4.0, 6.0), use d_{L1}(x, q) = $\sum_j |x_j - q_j|$ and cosine distance d_{cos} = 1 - $\frac{xq}{||x||\,||q||}$

L1 nearest: ID 7 (1.8, A), ID 6 (3.9, B), ID 5 (4.1, A) → Class A

Cosine nearest: ID 7 (0.0013, A), ID 3 (0.0146, B), ID 1 (0.0164, A) → Class A

4.

Using the training data, both classes have prior probability 4/8=0.5 For X = [Sunny, hot, Normal, False]:

Score(Yes) = $0.5*\frac{2}{4}*\frac{1}{4}*\frac{2}{4}*\frac{3}{4}$ = 0.02344

Score(No) = $0.5*\frac{1}{4}*\frac{1}{4}*\frac{2}{4}*\frac{2}{4}$ = 0.00781

Score(Yes)> Score(No), predict Yes

5.

The dataset has 6 Yes and 3 No:

H(S) = $-\frac{6}{9}log_2\frac{6}{9}-\frac{3}{9}log_2\frac{3}{9}$ = 0.7642

Information gains:

IG(Weather) = 0.4581 IG(Temperature) = 0.1409 IG(Wind) = 0.2248

Choose Weather as the root. Overcast and Rain subsets are all Yes. For the Sunny subset, splitting by Wind perfectly separates the labels:

Wind = High → No

Wind = Medium → Yes

6.

Start with Data Points 1 and 3 as centroids. After assigning each point to its nearest centroid in round 1:

Cluster 1: points{1, 2, 5, 7}, centroid

$(\frac{7.8}{4}, \frac{11.8}{4}, \frac{4.8}{4})$ = (1.95, 2.95, 1.2)

Cluster 2: points {3, 4, 6, 8}, centroid

$(\frac{31}{4}, \frac{27}{4}, \frac{35}{4})$ = (7.75, 6.75, 8.75)

Round 2 gives the same assignments and centroids.

7.

Conv1: 8(1*3*3)+8 = 80

Conv2: 16(8*3*3)+16 = 1168

Conv3: 32(16*3*3)+32 = 4640

Linear: 2048*10+10 = 20490

Total parameters: 80+1168+4640+20490 = 26378

8.

Assumption: standardize the entire matrix using population standard deviation, Z = (A-μ)/σ. $\mu = 8.52,\ \sigma \approx 4.859$ Using the CNN cross-correlation convention:

Conv(Z, K) $\approx \begin{bmatrix} -3.4987 & 1.2348 & 3.7045 \\ 0.6174 & 0.2058 & -0.2058 \\ 0.8232 & -1.6464 & -1.0290 \end{bmatrix}$

After ReLU and row-wise flattening:

V = [0, 1.2348, 3.7045, 0.6174, 0.2058, 0, 0.8232, 0, 0]

vW = 0.8232